

MindFlex AI

Technical Stack & Architecture Report

MindFlex AI - Tech Stack & Architecture Report

This report outlines the complete frontend, backend, database, APIs, and features utilized in MindFlex AI - Emotion-Aware Virtual Learning Assistant.

System Architecture Overview

MindFlex AI is designed as a Single-Process Web Application for maximum speed, easy deployment, and low execution overhead. It combines client-side interactivity, server-side Python logic, and hybrid local/cloud LLM APIs into a unified dashboard.

```
graph TD
    Client[Browser Frontend] -->|Interactive Actions| Streamlit[Streamlit Python Server]
    Streamlit -->|Local JSON Store| DB[users.json & study_planner.json]
    Streamlit -->|Native JS Speech| TTS[Browser Web Speech API]
    Streamlit -->|Hybrid Toggle| LLM{LLM Router}
    LLM -->|Local Inference| Ollama[Local Ollama / Qwen-2.5 3B]
    LLM -->|Cloud Inference| Groq[Groq API / Llama-3.3 70B]
```

1. Frontend Architecture (User Interface)

The frontend is built on Streamlit (Python), styled with customized raw CSS injects to deliver a dark, premium, glassmorphic layout.

A. Styling & Aesthetics

- Glassmorphic Theme: Utilizes HSL color tokens, semi-transparent backgrounds (`rgba`), backdrop blurs, and subtle neon border glows.
- Typography: Injects Google Fonts (*Plus Jakarta Sans*) globally to override standard browser fonts.
- Responsive Collapsibles: Customized expanders (`summary`/`details` selectors) using border-rotated CSS chevrons to prevent Broad Ligature overrides, ensuring clean layout rendering in offline environments.

B. UI Component Breakdown

- Unified Sign-In/Sign-Up Landing Page:
- Left Column: High-fidelity AI Capabilities grid (Emotion explanations, sentiment checks, speech synthesis, and concept labs).
- Right Column: Tabs-based credential cards (`Sign In` / `Sign Up`) with instant error validation prompts.
- Personalized Student Dashboard:
- Sidebar: Student welcome card, dynamic breathing pulse status avatar showing active tutor state (Focused, Confused, Engaged), state counter tracker, and a logout button.
- Learning Dashboard Tab:

- Subject/Topic selector dropdown.
 - Interactive accordions displaying incremental lesson walkthroughs.
 - Multi-subject interactive playground calculators (Algebra discriminant solver, Calculus variable substitution animator, Physics kinetic energy work metric slider).
 - Study Planner Tab:
 - Core metrics row displaying total logged topics, average score, mastered counts, and weak areas.
 - Schedule sub-tab: Generates a custom 3-day revision/learning plan.
 - My Topics sub-tab: Lists logged topic scores with individual progress meters.
 - Log Topic sub-tab: Form to record quiz outcomes (0-100) and update schedules.
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2. Backend & Integration Logic

The backend logic runs natively inside Streamlit's reactive execution thread.

A. Natural Language Processing (LLM Integrations)

The system supports a Dual-LLM configuration toggleable in real-time:

1. Local Ollama (100% Offline):
 - Model: ``qwen2.5:3b``
 - Connection: Direct local loopback request (``http://127.0.0.1:11434/api/chat``).
2. Groq Cloud (Ultra-Fast API):
 - Model: ``llama-3.3-70b-versatile``
 - Connection: Raw HTTPS POST request to Groq API endpoint (~1000 tokens/sec generation).
 - API Key Resolver: Checks Streamlit Cloud ``st.secrets`` first (ideal for live hosting), then falls back to local ``.env`` files or manual sidebar form input.

B. Real-Time Emotion Adaptive Prompting

- Frustration Parser: Automatic regex heuristics search chat prompts for words indicating confusion (``**stuck**``, ``**don't understand**``, ``**confused**``) or challenge (``**practice**``, ``**quiz**``).
- State Engine: Instantly changes session states:
 - Confused : Instructs the LLM to output simplified, analogy-driven, patient explanations, ending with simple checkpoint questions.
 - Focused : Instructs the LLM to write precise mathematical derivations and note tricky edge cases (e.g. constants of integration).
 - Engaged : Instructs the LLM to output a short explanation followed by a concrete practice problem for the student to solve.

C. Client-Side Speech Synthesis (TTS)

- API: Browser Web Speech API (``window.speechSynthesis``).
 - Implementation: Uses a hidden Streamlit HTML component (``st.components.v1.html``) containing an iframe JavaScript block.
 - Play/Stop Toggle: Generates custom ``<button>`` (Speak) and ``<button>`` (Cancel) control loops. Triggers ``speechSynthesis.cancel()`` instantly to stop audio or overlaying tracks when switching buttons, resetting chats, or entering new questions.
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3. Database & Data Persistence

The application maintains state and user progress locally without requiring heavy external database services (ideal for offline evaluations):

1. User Database (`users.json`):

- File-based local JSON repository.
- Stores credentials securely as key-value pairs (lowercase username to password mapping).

2. Planner Database (`study\_planner.json`):

- Stores personal topic coverage logs for each student.
- Format:

```
{
  "username": {
    "topics": [
      {
        "id": 1718384210,
        "name": "Integration by Substitution",
        "score": 85,
        "date": "2026-06-14"
      }
    ],
    "weak_areas": []
  }
}
```

4. Hackathon Team & Roles

- Jay Gaikar ([@jaygaikar-09](https://github.com/jaygaikar-09)): Lead Developer, Streamlit & UI design, state management, hybrid LLM routing.
- Somya Asati ([@somyaasati12-del](https://github.com/somyaasati12-del)): Frontend UI layouts, capabilities list, initial concept styling.
- Vaishnavi Asati ([@vaishnaviasati21](https://github.com/vaishnaviasati21)): Study Planner feature design, schedule algorithms, and state parameters.
- Ashish Pul ([@ashishpul](https://github.com/ashishpul)): Contributor, documentation structure, and repository guidelines.